

Answer **all** questions from this paper.

1	(a)	A swimmer is swimming at a constant speed in a pool. Drag forces due to the water oppose the motion of the swimmer. Explain why the swimmer travels at constant speed. [2]															
		Solution: The swimmer experiences a forward force from the water (thrust) which is equal and opposite to the drag force so that the net force is zero. 1 According to Newton's first law of motion, since there is no net force the swimmer will travel at a constant speed. 1															
	(b)	The power output P of a swimmer used to overcome the drag forces travelling when at speed v is given by $P = \frac{1}{2} C_D \rho A v^3$ where C_D is the drag coefficient, ρ is the density of water and A is the frontal area of the swimmer. In an experiment to measure the C_D for freestyle, the data for a particular swimmer is collected. The data is shown in Table 1.1. Table 1.1 <table border="1" data-bbox="343 1431 1388 1691"> <thead> <tr> <th>quantity</th><th>magnitude</th><th>uncertainty</th></tr> </thead> <tbody> <tr> <td>P/W</td><td>294</td><td>± 2</td></tr> <tr> <td>$\rho/\text{kg m}^{-3}$</td><td>1000</td><td>± 1</td></tr> <tr> <td>A/m^2</td><td>0.20</td><td>± 0.01</td></tr> <tr> <td>$v/\text{m s}^{-1}$</td><td>1.4</td><td>± 0.1</td></tr> </tbody> </table> Determine the C_D of the swimmer, with its actual uncertainty. Give your answer to an appropriate number of significant figures.	quantity	magnitude	uncertainty	P/W	294	± 2	$\rho/\text{kg m}^{-3}$	1000	± 1	A/m^2	0.20	± 0.01	$v/\text{m s}^{-1}$	1.4	± 0.1
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			$C_D = \dots\dots\dots \pm \dots\dots\dots$ [4]
		Solution: $C_D = \frac{2P}{\rho A v^3} = \frac{2(294)}{1000(0.20)(1.4)^3}$ $C_D = 1.07$ $\frac{\Delta C_D}{C_D} = \frac{\Delta P}{P} + \frac{\Delta \rho}{\rho} + \frac{\Delta A}{A} + 3 \frac{\Delta v}{v} = \frac{2}{294} + \frac{1}{1000} + \frac{0.01}{0.20} + 3 \frac{0.1}{1.4}$ $\frac{\Delta C_D}{C_D} = 0.272$ $\Delta C_D = 0.272 \times 1.07 \approx 0.3$ $C_D = 1.1 \pm 0.3$	1 1 1 1
	(c)	(i) Derive, from the definition of power, an expression of the drag force F experienced by the swimmer in terms of the velocity v .	[2]
		Solution: Since $P = \frac{W}{t} = \frac{Fs}{t} = Fv$ $F = \frac{P}{v} = \frac{\frac{1}{2} C_D \rho A v^3}{v}$ $F = \frac{1}{2} C_D \rho A v^2$	1 1
		(ii) Hence or otherwise, calculate the work done by the swimmer to overcome the drag force when he swims a distance of 50 m at a constant speed of 1.4 m s ⁻¹ .	

			work done = J	[2]
			Solution: $W = Fs = \frac{1}{2} C_D \rho A v^2 s = \frac{1}{2} (1.07)(1000)(0.20)(1.4)^2 (50)$ $W = 10500 \text{ J}$	1 1